Automatic Weapon Detection and Recognition using Deep Learning.

Abstract

Real-time object detection to improve surveillance methods is one of the sought -after applications of Convolutional Neural Networks (CNNs). This research work has approached the detection of handguns in areas monitored by cameras. Gun violence and mass shootings are also on the rise in certain parts of the world. Such incidents are time sensitive and can cause a huge loss to life and property. Hence, the proposed work has built a deep learning model based on the YOLOv3 algorithm that processes a video frame-by-frame to detect such anomalies in real-time and generate an alert for the concerned authorities.

Key Words: Convolutional Neural Networks (CNN), Surveillance methods, YOLOv3 Algorithm

Group Name: G-15

Group Members:

245622733002: ADDAGULLA ISHIKA

245622733303: CHOLLETI SRI VARSHINI

245622733304: GOPU SREEJA

Project Guide : Dr V.HARIKA

Signature of the guide